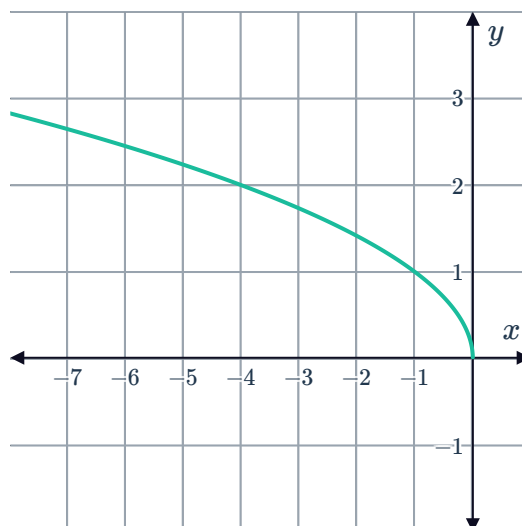


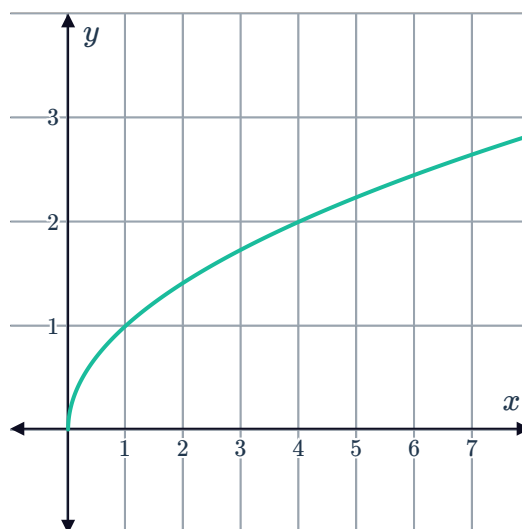
Key features



- 1 Consider the graph of $y = \sqrt{-x}$:
As x approaches $-\infty$, what does y approach?



- 2 Consider the graph of $y = \sqrt{x}$:
- a Describe the rate of increase of the function as x increases.
 - b State the axes intercepts.
 - c Does the function have an asymptote?
 - d Does the function have a limiting value?
 - e As x increases, what does y approach?

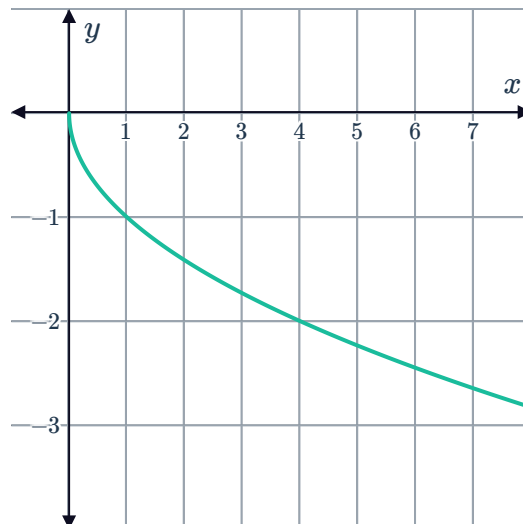


3 Consider the function $y = -\sqrt{x}$.



- a Complete the table of values. Round any values to two decimal places if necessary.
- b Can the function values ever be positive?
- c The graph of the function $y = -\sqrt{x}$ is shown. Is $y = -\sqrt{x}$ an increasing function or a decreasing function?
- d Describe the rate of decrease of the function as x increases.

x	0	1	2	3	4	5	9
y							

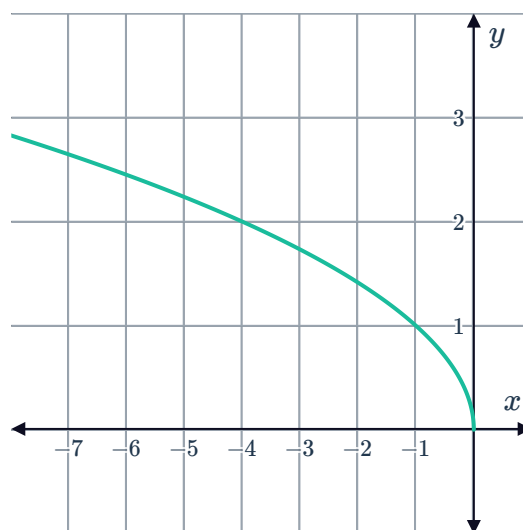


4 Consider the function $y = \sqrt{-x}$.

- a Complete the table of values. Round any values to two decimal places if necessary.

x	-5	-4	-3	-2	-1	0
y						

- b The graph of $y = \sqrt{-x}$ is given. Is $y = \sqrt{-x}$ an increasing function or a decreasing function?



5 Consider the function $y = \sqrt{x} + 3$.

- a Can y ever be negative?
- b As x gets larger and larger, what value does y approach?
- c Determine the y -intercept of the curve.
- d How many x -intercepts does it have?
- e Sketch the graph.

6 Consider the function $y = 2\sqrt{x} + 3$.



- a Is the function increasing or decreasing from left to right?
- b Is the function more or less steep than $y = \sqrt{x}$?
- c What are the coordinates of the vertex?
- d Sketch the graph.

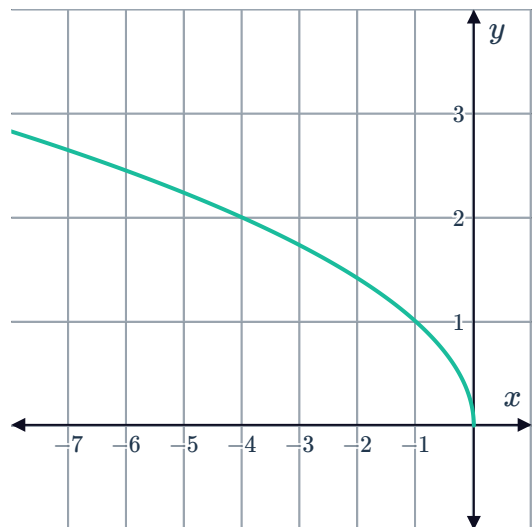
7 Consider the function $y = -\frac{1}{2}\sqrt{x} + 2$.

- a Is the function increasing or decreasing from left to right?
- b Is the function more or less steep than $y = \sqrt{x}$?
- c What are the coordinates of the vertex?
- d Sketch the graph.

Domain and range

8 Consider the function $y = \sqrt{-x}$.

- a State the domain of the function.
- b State the range of the function.



9 Consider the function $y = \sqrt{x}$.

- a Complete the table of values. Round any values to two decimal places if necessary.
- b State the domain of the function.
- c State the range of the function.
- d As x gets larger and larger, what value does y approach?
- e Sketch the graph of $y = \sqrt{x}$.

x	0	1	2	3	4	5	9
y							

10 Consider the function $y = -\sqrt{x}$.



a State the domain of the function.

b State the range of the function.

11 The function $y = \sqrt{x}$ has domain $x \geq 0$ and range $y \geq 0$.
What is the domain and range of $y = \sqrt{x} - 2$?

12 Consider the function $y = \sqrt{x - 5}$.

a State the domain of the function.

b State the range of the function.

c Do the functions $y = \sqrt{x}$ and $y = \sqrt{x - 5}$ increase at the same rate?

13 Consider the function $y = \sqrt{-x} + 6$.

a What is the smallest possible function value?

b State the domain of the function.

c State the range of the function.

14 A square root function has a range of $y \leq 0$ and a domain of $x \geq 0$. Determine whether the following could be the equation of the function:

a $y = \sqrt{x}$

b $y = -\sqrt{-x}$

c $y = -\sqrt{x}$

d $y = 5\sqrt{x}$

e $y = -5\sqrt{x}$

f $y = \sqrt{-x}$

15 For each of the following functions:

i Sketch the graph

ii State the domain

iii State the range

a $f(x) = -\sqrt{x + 1}$

b $f(x) = -2\sqrt{x + 5}$

c $f(x) = 3\sqrt{\left(\frac{1}{3}x\right)}$

d $f(x) = -\frac{\sqrt{x}}{2} - 2$

e $f(x) = \frac{\sqrt{x - 1}}{2} + 2$

16 For each of the following functions:

i State the domain of the function.

ii State the range of the function.

iii Sketch the graph.

a $u = -\sqrt{x} + 5$

b $u = -5\sqrt{x}$

c $u = -\sqrt{-x}$

d $u = \sqrt{x - 3} + 2$



17 For which values of x do the following expressions evaluate to a real number?

a $\sqrt{7x}$

b $\sqrt{x-2}$

c $\sqrt{3-x}$

d $\sqrt{15-5x}$

e $\sqrt{x^2+6}$

18 Consider the function $f(x) = \sqrt{x-2} + 3$. State the domain of the function using interval notation.

Transformations

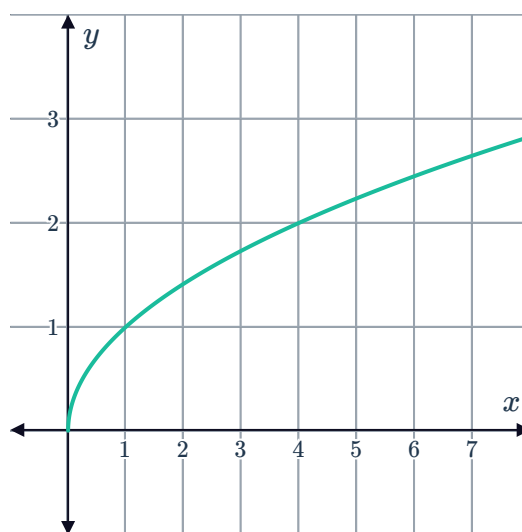
19 The graph of $y = \sqrt{x}$ has a vertex at $(0,0)$. By considering the transformations that have taken place, state the coordinates of the vertex of $y = -\sqrt{x} + 3$.

20 The graph of $y = \sqrt{x}$ has been translated to the graph of $y = \sqrt{x} - 4$.

- a** Describe the transformation that has occurred on the original function.
- b** Hence, sketch the graph of $y = \sqrt{x} - 4$.

21 Consider the graph of $y = \sqrt{x}$ shown:

- a**
 - i** Sketch the curve after $y = \sqrt{x}$ has been reflected about the y -axis.
 - ii** What is the equation of this new graph?
- b**
 - i** Sketch the curve after $y = \sqrt{x}$ has been reflected about the x -axis.
 - ii** What is the equation of this new graph?



22 Consider the function $y = \sqrt{x}$:

- a** Describe how we can transform the graph of $y = \sqrt{x}$ to get the graph of $y = \sqrt{x-4} + 3$.
- b** Hence, sketch the graph of $y = \sqrt{x-4} + 3$.

23 Sketch the curve $y = 3\sqrt{x-2} + 3$.